Human review packet: Roth82StableMatching

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: Roth82StableMatching
Paper id: `Roth82StableMatching`
Source version: Mathematics of Operations Research 7(4), 1982, printed pages 617-628
Source record: `audit/paper_statement_map.json`
Rows: 21
Generated: 2026-08-18
Source URL: official source

Review status: 0/21 source-claim screens; 0/17 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper Roth82StableMatching --serve
```

Open `http://127.0.0.1:8765/` in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

Contents

- theorem3_no_stable_truthful_procedureSpec
- theorem4_efficient_truthful_procedure_existsSpec
- theorem5_optimal_side_truthfulSpec
- corollary5_1_first_choice_need_not_be_misrepresentedSpec
- lemma1_simple_report_same_partnerSpec
- lemma2_simple_report_harms_no_manSpec
- theorem6_no_outcome_strictly_better_for_all_menSpec
- theorem7_later_choice_manipulation_unavoidableSpec
- manReportStrictlyRanksPartnerFirst_iff_source_definitionSpec
- matchingProcedure_eq_source_definitionSpec
- sideOptimalStableOutcome_iff_source_definitionSpec
- sourceMarriageModel_iff_source_definitionSpec

Material library prerequisites

`Assignment M W`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_1 → Type u_2 → Type (max u_1 u_2)
```

Exact Lean library declaration

```
/-- A matching between Men and Women. -/
structure Assignment (M W : Type*) where
  m_match : M → Option W
  w_match : W → Option M
  consistent_m : ∀ m w, m_match m = some w ↔ w_match w = some m
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.Assignment.mk`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{M : Type u_1} →
  {W : Type u_2} →
    (m_match : M → Option W) →
      (w_match : W → Option M) →
        (∀ (m : M) (w : W), m_match m = some w ↔ w_match w = some m) → EconCSLib.Matching.Assignment M W
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.Assignment.swap

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_1} →
  {W : Type u_2} →
    (mu : EconCSLib.Matching.Assignment M W) → { m_match := mu.w_match, w_match := mu.m_match, consistent_m := ⋯ }
```

Exact Lean library declaration

```
def swap {M W : Type*} (mu : Assignment M W) : Assignment W M where
  m_match := mu.w_match
  w_match := mu.m_match
  consistent_m w m := (mu.consistent_m m w).symm
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.DAState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
Type u_3 → Type u_4 → Type (max u_3 u_4)
```

Exact Lean library declaration

```
structure DAState (M W : Type*) where
  m_match : M → Option W
  w_match : W → Option M
  m_proposals : M → Finset W
  consistent : ∀ m w, m_match m = some w ↔ w_match w = some m
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.DAState.m_proposals`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_3} → {W : Type u_4} → (self : EconCSLib.Matching.DAState M W) → (a : M) → self.3 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.BestRemainingWoman

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (s : EconCSLib.Matching.DAState M W) (m : M) (w : W),
  w ∈ s.m_proposals m ∧ 0 ≤ val_m m w ∧ ∀ w' ∈ s.m_proposals m, 0 ≤ val_m m w' → val_m m w' ≤ val_m m w
```

Exact Lean library declaration

```
def BestRemainingWoman (val_m : M → W → ℝ) (s : DAState M W) (m : M) (w : W) : Prop :=
  w ∈ s.m_proposals m ∧ 0 ≤ val_m m w ∧
  ∀ w' ∈ s.m_proposals m, 0 ≤ val_m m w' → val_m m w' ≤ val_m m w
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.DAState.m_match`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_3} → {W : Type u_4} → (self : EconCSLib.Matching.DAState M W) → (a : M) → self.1 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.DAState.w_match

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_3} → {W : Type u_4} → (self : EconCSLib.Matching.DAState M W) → (a : W) → self.2 a
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.DAState.consistent`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {M : Type u_3} {W : Type u_4} (self : EconCSLib.Matching.DAState M W) (m : M) (w : W),
  self.m_match m = some w ↔ self.w_match w = some m
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.DAState.mk

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
{M : Type u_3} →
  {W : Type u_4} →
    (m_match : M → Option W) →
      (w_match : W → Option M) →
        (M → Finset W) → (∀ (m : M) (w : W), m_match m = some w ↔ w_match w = some m) → EconCSLib.Matching.DAState M W
```

Exact Lean library declaration

```
library declaration is not registered or discoverable
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.IsActiveMan

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
∀ {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (s : EconCSLib.Matching.DAState M W) (m : M),
  s.m_match m = none ∧ ∃ w ∈ s.m_proposals m, 0 ≤ val_m m w
```

Exact Lean library declaration

```
def IsActiveMan (val_m : M → W → ℝ) (s : DAState M W) (m : M) : Prop :=
  s.m_match m = none ∧ ∃ w ∈ s.m_proposals m, 0 ≤ val_m m w
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.exists_best_woman`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

```
∀ {M : Type u_1} {W : Type u_2} [Fintype M] [Fintype W] [DecidableEq M] [DecidableEq W] (val_m : M → W → ℝ)
  (s : EconCSLib.Matching.DAState M W) (m : M),
  EconCSLib.Matching.IsActiveMan val_m s m → ∃ w, EconCSLib.Matching.BestRemainingWoman val_m s m w
```

Exact Lean library declaration

```
lemma exists_best_woman (val_m : M → W → ℝ) (s : DAState M W) (m : M)
    (hact : IsActiveMan val_m s m) : ∃ w, BestRemainingWoman val_m s m w
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.removeProposal`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_1} →
  {W : Type u_2} →
    [inst : DecidableEq M] →
      [inst_1 : DecidableEq W] →
        (s : EconCSLib.Matching.DAState M W) →
          (m : M) → (w : W) → (a : M) → if a = m then s.m_proposals m \ {w} else s.m_proposals a
```

Exact Lean library declaration

```
def removeProposal (s : DAState M W) (m : M) (w : W) : M → Finset W :=
  fun m' => if m' = m then s.m_proposals m \ {w} else s.m_proposals m'
```

Recorded source-to-library screening   Verdict: `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.daStep

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_1} →
  {W : Type u_2} →
    [inst : Fintype M] →
      [inst_1 : Fintype W] →
        [inst_2 : DecidableEq M] →
          [inst_3 : DecidableEq W] →
            (val_m : M → W → ℝ) →
              (val_w : W → M → ℝ) →
                (s : EconCSLib.Matching.DAState M W) →
                  have x := Classical.propDecidable (∃ m, EconCSLib.Matching.IsActiveMan val_m s m);
                  if h : ∃ m, EconCSLib.Matching.IsActiveMan val_m s m then
                    let m := Classical.choose h;
                    have hact := ⋯;
                    have w_exists := ⋯;
                    let w := Classical.choose w_exists;
                    have hw_best := ⋯;
                    have new_proposals := EconCSLib.Matching.removeProposal s m w;
                    let m_current := s.w_match w;
                    have accepts :=
                      match m_current with
                      | none => 0 ≤ val_w w m
                      | some m' => val_w w m' < val_w w m;
                    have x := Classical.propDecidable accepts;
                    if hacc : accepts then
                      let new_w_match := Function.update s.w_match w (some m);
                      let new_m_match := fun m'' =>
                        if m'' = m then some w else if m_current = some m'' then none else s.m_match m'';
                      { m_match := new_m_match, w_match := new_w_match, m_proposals := new_proposals, consistent := ⋯ }
                    else { m_match := s.m_match, w_match := s.w_match, m_proposals := new_proposals, consistent := ⋯ }
                  else s
```

Exact Lean library declaration

```
noncomputable def daStep (val_m : M → W → ℝ) (val_w : W → M → ℝ)
    (s : DAState M W) : DAState M W :=
  have _ : Decidable (∃ m, IsActiveMan val_m s m) := Classical.propDecidable _
  if h : ∃ m, IsActiveMan val_m s m then
    let m := Classical.choose h
    let hact := Classical.choose_spec h
    let w_exists := exists_best_woman val_m s m hact
    let w := Classical.choose w_exists
    let hw_best := Classical.choose_spec w_exists
    let new_proposals := removeProposal s m w
    let m_current := s.w_match w
    let accepts :=
      match m_current with
      | none => 0 ≤ val_w w m
      | some m' => val_w w m' < val_w w m
    have _ : Decidable accepts := Classical.propDecidable _
    if hacc : accepts then
      let new_w_match := Function.update s.w_match w (some m)
      let new_m_match := fun m'' =>
        if m'' = m then some w
        else if m_current = some m'' then none
        else s.m_match m''
      { m_match := new_m_match
        w_match := new_w_match
        m_proposals := new_proposals
        consistent := by
          simpa [new_m_match, new_w_match, m_current] using
            acceptMatch_consistent s hact.1 }
    else
      { s with m_proposals := new_proposals }
  else
    s
```

Recorded source-to-library screening   **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.initialDAState`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(M : Type u_3) →
  (W : Type u_4) →
    [inst : Fintype W] →
      { m_match := fun x => none, w_match := fun x => none, m_proposals := fun x => Finset.univ, consistent := ⋯ }
```

Exact Lean library declaration

```
def initialDAState (M W : Type*) [Fintype W] : DAState M W where
  m_match _ := none
  w_match _ := none
  m_proposals _ := Finset.univ
  consistent m w := by simp
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Matching.deferredAcceptanceState

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_1} →
  {W : Type u_2} →
    [inst : Fintype M] →
      [inst_1 : Fintype W] →
        [inst_2 : DecidableEq M] →
          [inst_3 : DecidableEq W] →
            (val_m : M → W → ℝ) →
              (val_w : W → M → ℝ) →
                have max_steps := Fintype.card M * Fintype.card W;
                List.foldl (fun s x => EconCSLib.Matching.daStep val_m val_w s) (EconCSLib.Matching.initialDAState M W)
                  (List.range max_steps)
```

Exact Lean library declaration

```
noncomputable def deferredAcceptanceState (val_m : M → W → ℝ) (val_w : W → M → ℝ) : DAState M W :=
  let max_steps := Fintype.card M * Fintype.card W
  (List.range max_steps).foldl (fun s _ => daStep val_m val_w s) (initialDAState M W)
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

`EconCSLib.Matching.deferredAcceptance`

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{M : Type u_1} →
  {W : Type u_2} →
    [inst : Fintype M] →
      [inst_1 : Fintype W] →
        [inst_2 : DecidableEq M] →
          [inst_3 : DecidableEq W] →
            (val_m : M → W → ℝ) →
              (val_w : W → M → ℝ) →
                { m_match := (EconCSLib.Matching.deferredAcceptanceState val_m val_w).m_match,
                  w_match := (EconCSLib.Matching.deferredAcceptanceState val_m val_w).w_match, consistent_m := ⋯ }
```

Exact Lean library declaration

```
noncomputable def deferredAcceptance (val_m : M → W → ℝ) (val_w : W → M → ℝ) : Assignment M W where
  m_match := (deferredAcceptanceState val_m val_w).m_match
  w_match := (deferredAcceptanceState val_m val_w).w_match
  consistent_m := (deferredAcceptanceState val_m val_w).consistent
```

Recorded source-to-library screening **Verdict:** `not recorded`

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 130-138

Verbatim source input

```
defined on W. When no confusion will occur, P(mi) will sometim

thus the statement "wk is preferred by mi to w." can be written as w

Each wi in W has a similar preference P(wi) defined on

= (P(m,), . . ., P(wn)) the 2n-vector of all the agents' preferenc

times be referred to as the preference profile.
```

Expanded PaperInterface specification

```
∀ (M : Type u) (W : Type v),
  Roth82StableMatching.PaperInterface.preferenceProfile M W =
    { profile //
      (∀ (m : M) (w w' : W), profile.1 m w = profile.1 m w' → w = w') ∧
        ∀ (w : W) (m m' : M), profile.2 w m = profile.2 w m' → m = m' }
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.preferenceProfile_eq_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 2

Source locator: cited publication, lines 140-145

Verbatim source input

```
An outcome of the (monogamous) marriage problem is a onemen and women, i.e., an invertible function x: M - W. An outco

by x = [(ml,x(ml)),(m2,x(m2)),. .. , (mn,x(mM))] where x(mi

matched with man m,, and x- (wj) = mi is the man matched wit
A matching x is stable if no man and no woman who are not m
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} (mu : EconCSLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.completeMarriageOutcome mu ↔
    (∀ (m : M), ∃ w, mu.m_match m = some w) ∧ ∀ (w : W), ∃ m, mu.w_match w = some m
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.completeMarriageOutcome_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 3

Source locator: cited publication, lines 145-165

Verbatim source input

```
A matching x is stable if no man and no woman who are not m

at x prefer each other to their partners at x. That is, x is stable if t

and woman Wq such that both

(i)

and

WqPkx(mk)

(ii) mkPqX l(wq).

If a pair mk and wq do exist satisfying (i) a

and wq. The motivation for this terminol
that the set of stable outcomes is equal
results if any man and woman may mar
agent's preference for an outcome is dete
partners).
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ) (mu : EconCSLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu ↔
    Roth82StableMatching.PaperInterface.completeMarriageOutcome mu ∧
      ¬∃ m w,
          (match mu.m_match m with
              | none => 0
              | some w' => val_m m w') <
              val_m m w ∧
            (match mu.w_match w with
              | none => 0
              | some m' => val_w w m') <
              val_w w m
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.sourceStableMarriage_iff_source_definition
```

Recorded source-to-Spec screening Verdict: `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 4

Source locator: cited publication, lines 167-175

Verbatim source input

```
The marriage problem as outlined above differs from the general matching problem
in three principal respects. First, in the marriage problem, each agent is to be matched

with exactly one partner, but in the general matching problem, different agents may
need to be matched with different numbers of partners, so that each agent has a
"quota," and an outcome is a function which matches each agent with his quota of
partners.6 For our purposes, this difference between the marriage problem and the
general matching problem turns out to be of no consequence, since all the arguments
which will be used can be carried over virtually unchanged to the general case. Second,
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} (quotaM : M → ℕ) (quotaW : W → ℕ)
  (mu : Roth82StableMatching.PaperInterface.ManyToManyQuotaAssignment M W),
  Roth82StableMatching.PaperInterface.fillsIndividualQuotas quotaM quotaW mu ↔
    (∀ (m : M), (mu.partnersM m).Finite ∧ (mu.partnersM m).ncard = quotaM m) ∧
      ∀ (w : W), (mu.partnersW w).Finite ∧ (mu.partnersW w).ncard = quotaW w
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.fillsIndividualQuotas_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 313-323

Verbatim source input

```
A different kind of problem concerning incentives arises for matching procedures
which do not result in stable outcomes, since such procedures give at least one pair of
agents the incentive to disregard the matching procedure, and seek an alternative
outcome. Of course, it may be possible to compel the agents to accept the outcome, in
spite of these incentives. For instance, the procedure by which some high school
athletes are matched with colleges involves signing "letters of intent," after which
athletes are effectively prohibited from further negotiating with other schools. (Professional athletes in several sports are
↪  matched with teams under an even more restrictive
draft, which prevents a player from negotiating with any team but the one which drafts
him.) However in situations in which compulsion plays no part, it is desirable for a
matching procedure to yield stable outcomes. A procedure which does so for arbitrary
preference profiles will be called a stable matching procedure.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} (mechanism : (M → W → ℝ) → (W → M → ℝ) → EconCSLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure mechanism ↔
    ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
      Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
        Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w (mechanism val_m val_w)
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 340-351

Verbatim source input

```
Before proceeding further with this discussion, we need to make precise what is
meant by a procedure which gives agents an incentive to reveal their true preferences.
Once a given mechanism for aggregating stated preferences is adopted, the matching
problem becomes a noncooperative game among the agents, whose payoff consists of
the outcome which results, and whose strategy choices consist of what preferences to
state (or act according to, cf. footnote 9). Here we define a procedure which gives the
agents the incentives to reveal their true preferences as one which aggregates stated

preferences in a manner such that, in the resulting noncooperative game, it is a
dominant strategy for each player to state his preferences honestly.10 In such a
procedure, no matter what preferences other players may state, a player who misrepresents his preferences can achieve no better
↪  outcome than if he had stated them
correctly, and he may, of course, do worse."
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : DecidableEq M] [inst_1 : DecidableEq W]
  (mechanism : (M → W → ℝ) → (W → M → ℝ) → EconCSLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.truthfulForAllAgents mechanism ↔
    (∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
        Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
          ∀ (m : M) (report_m : W → ℝ),
            Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
              (match (mechanism (Function.update val_m m report_m) val_w).m_match m with
                | none => 0
                | some w => val_m m w) ≤
                match (mechanism val_m val_w).m_match m with
                | none => 0
                | some w => val_m m w) ∧
      ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
        Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
          ∀ (w : W) (report_w : M → ℝ),
            Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
              (match (mechanism val_m (Function.update val_w w report_w)).w_match w with
                | none => 0
                | some m => val_w w m) ≤
                match (mechanism val_m val_w).w_match w with
                | none => 0
                | some m => val_w w m
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.truthfulForAllAgents_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 7

Source locator: cited publication, lines 433-435

Verbatim source input

```
To see the role played by stability in Theorem 3, observe that there are effici
matching procedures, i.e., procedures which always yield Pareto optimal (but not
necessarily stable) outcomes, which do not give players any incentive to misrepresent
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} (mechanism : (M → W → ℝ) → (W → M → ℝ) → EconCSLib.Matching.Assignment M W),
  Roth82StableMatching.PaperInterface.efficientMatchingProcedure mechanism ↔
    ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
      Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
        Roth82StableMatching.PaperInterface.paretoOptimal val_m val_w (mechanism val_m val_w)
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.efficientMatchingProcedure_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 227

Verbatim source input

```
THEOREM 1. The set of stable outcomes is always nonempty.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [DecidableEq M] [DecidableEq W]
  (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Fintype.card M = Fintype.card W →
    Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
      ∃ mu, Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.theorem1_stable_outcome_exists
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 258-259

Verbatim source input

```
THEOREM 2. There is a stable outcome weakly preferred by every man to any other
stable outcome, and one weakly preferred by every woman.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [DecidableEq M] [DecidableEq W]
  (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Fintype.card M = Fintype.card W →
    Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
      (∃ mu,
          Roth82StableMatching.PaperInterface.menOptimal val_m val_w mu ∧
            Roth82StableMatching.PaperInterface.completeMarriageOutcome mu) ∧
        ∃ nu,
          Roth82StableMatching.PaperInterface.womenOptimal val_m val_w nu ∧
            Roth82StableMatching.PaperInterface.completeMarriageOutcome nu
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.theorem2_side_optimal_stable_outcomes
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 356-357

Verbatim source input

```
THEOREM 3. No stable matching procedure for the general matching problem exists
for which truthful revelation of preferences is a dominant strategy for all agents.
```

Expanded PaperInterface specification

```
¬∃ mechanism,
    Roth82StableMatching.PaperInterface.sourceStableMatchingProcedure mechanism ∧
      Roth82StableMatching.PaperInterface.truthfulForAllAgents mechanism
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.theorem3_no_stable_truthful_procedure
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 11

Source locator: cited publication, lines 438-439

Verbatim source input

```
THEOREM 4. Efficient matching procedures exist for which truthful revelation of
preferences is a dominant strategy for every agent.
```

Expanded PaperInterface specification

```
∀ (n : ℕ),
  ∃ mechanism,
    Roth82StableMatching.PaperInterface.efficientMatchingProcedure mechanism ∧
      Roth82StableMatching.PaperInterface.truthfulForAllAgents mechanism
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.theorem4_efficient_truthful_procedure_exists
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 468-475

Verbatim source input

```
THEOREM 5. In the matching procedure which always yields the optimal stable

outcome for a given one of the two sets of agents (i.e., for M or for W), truthful revelation
is a dominant strategy for all the agents in that set.


[form-feed in source extraction]
624

ALVIN E. ROTH
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
  [inst_3 : DecidableEq W] (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Fintype.card M = Fintype.card W →
    Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
      (Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w
            (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w) ∧
          ∀ (m : M) (report_m : W → ℝ),
            Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
              (match
                  (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m)
                        val_w).m_match
                    m with
                | none => 0
                | some w => val_m m w) ≤
                match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
                | none => 0
                | some w => val_m m w) ∧
        Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w
            (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w) ∧
          ∀ (w : W) (report_w : M → ℝ),
            Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
              (match
                  (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m
                        (Function.update val_w w report_w)).w_match
                    w with
                | none => 0
                | some m => val_w w m) ≤
                match (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w).w_match w with
                | none => 0
                | some m => val_w w m
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.theorem5_optimal_side_truthful
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 477-535

Verbatim source input

```
COROLLARY 5.1. In the matching procedure which always yields the optimal stable
outcome for a given set of agents, the agents in the other set have no incentive to
misrepresent their first choice.

Note that both results are phrased in terms of "the" matching procedure which
yields a particular outcome. There are obviously different procedures which yield the
same outcome, but from the point of view of incentivies such procedures are equivalent, and can be regarded as a single
↪  procedure. Note also that Theorem 5 implies that,
although an agent can in general change the set of stable outcomes by misrepresenting
his preferences, no agent can manipulate his preferences in such a way that he prefers
his best outcome in the altered set of stable outcomes to his best outcome in the

original set of stable outcomes. The next section is devoted to the proof of thes
results, which is somewhat more complex than the proof of the earlier results.
presents some additional results about the structure of the set of stable outcom

which arise in the course of the proofs.

5. Proofs of Theorem 5 and its corollary. In this section it will be shown that t
repeated proposal procedure G used in the proof of Theorem 1 has the propertie
stated in Theorem 5 and Corollary 5.1. As discussed in section 2, to establish the
results for the general matching problem with strict preferences, it will be sufficient

demonstrate them for the marriage problem. Throughout this section, therefore, t
sets of agents will be M = { m .. . . , m} and W = (w, . . . , w }, the true preferenc
of the players will be given by the arbitrary preference profile P, and x = g(P) w
denote the outcome which results from the repeated proposal procedure when the true
preferences are stated.

To prove Theorem 5, we need to show that truthful revelation is a dominant strategy

for each mi in M. Since the (true) preference profile P is arbitrary, it will be sufficien
to show that, if P' is a preference profile which differs from P only in that P'(mi), say

replaces man m,'s true preferences P(mi), then man mi doesn't prefer the outco
y = g(P') to x = g(P). That is, we need to show that no successful misrepresentati

of preferences is possible, where a misrepresentation P'(mi) is defined to be

successful misrepresentation by man mi if y(mi)P(mi)x(mi). That is, a misrepres
tation P'(mi) is successful if mi prefers (according to his true preferences) the partner

y(m,) he's matched with when he misrepresents his preferences to the partner x(m
he's matched with when he states his true preferences. (Throughout this section
y = g(P') will denote an outcome resulting from misrepresentation by mi.)

We first show that only a certain kind of simple misrepresentation need

considered, since if any successful manipulation is possible, then it can be achieved
a simple misrepresentation. Specifically, if P'(mi) is an arbitrary misrepresentation
discussed above, then an equivalent simple misrepresentation P"(mi) is one in whic

y(mi)P"(mi)wj for all wj #y(mi). That is, P"(mi) is a preference relation which ha
y(m,) as the most preferred match of mi,. The justification for calling P"(mi) an
equivalent misrepresentation to P'(mi) is given by the following lemma, which sta

that mi will end up matched to the same partner, y(m), whether he misrepresents usin

P'(mi) or P"(mi). (The preference profile P", of course, denotes the one which diffe
from P only in that P"(mi,) replaces P(mi).
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
  [inst_3 : DecidableEq W] [Nonempty M] [Nonempty W],
  Fintype.card M = Fintype.card W →
    (∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
        Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
          Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w
              (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w) ∧
            ∀ (w : W) (report_w : M → ℝ),
              Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w) →
                ∃ report_w',
                  Roth82StableMatching.PaperInterface.strictMarriageDomain val_m (Function.update val_w w report_w') ∧
                    (∀ (mstar : M),
                        (∀ (m : M), m ≠ mstar → val_w w m < val_w w mstar) →
                          ∀ (m : M), m ≠ mstar → report_w' m < report_w' mstar) ∧
                      (match
                          (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m
                                (Function.update val_w w report_w)).w_match
                            w with
                        | none => 0
                        | some m => val_w w m) ≤
                        match
                          (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m
                                (Function.update val_w w report_w')).w_match
                            w with
                        | none => 0
```

```
                      | some m => val_w w m) ∧
      ∀ (val_m : M → W → ℝ) (val_w : W → M → ℝ),
        Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
          Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w
              (Roth82StableMatching.PaperInterface.womenDeferredAcceptance val_m val_w) ∧
            ∀ (m : M) (report_m : W → ℝ),
              Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m) val_w →
                ∃ report_m',
                  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m report_m') val_w ∧
                    (∀ (wstar : W),
                        (∀ (w : W), w ≠ wstar → val_m m w < val_m m wstar) →
                          ∀ (w : W), w ≠ wstar → report_m' w < report_m' wstar) ∧
                      (match
                          (Roth82StableMatching.PaperInterface.womenDeferredAcceptance
                                (Function.update val_m m report_m) val_w).m_match
                            m with
                        | none => 0
                        | some w => val_m m w) ≤
                        match
                          (Roth82StableMatching.PaperInterface.womenDeferredAcceptance
                                (Function.update val_m m report_m') val_w).m_match
                            m with
                        | none => 0
                        | some w => val_m m w
```

Lean proof endpoint (built separately)

`Roth82StableMatching.PaperInterface.corollary5_1_first_choice_need_not_be_misrepresented`

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 537

Verbatim source input

```
LEMMA 1. If y = g(P') and z = g(P") then z(mi) = y(mi).
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
  [inst_3 : DecidableEq W] (val_m : M → W → ℝ) (val_w : W → M → ℝ) (m : M) (report_m simple_report_m : W → ℝ)
  (wstar : W),
  Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m simple_report_m) val_w →
    (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m) val_w).m_match m =
        some wstar →
      (∀ (w : W), w ≠ wstar → simple_report_m w < simple_report_m wstar) →
        (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
                val_w).m_match
            m =
          (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m report_m) val_w).m_match m
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.lemma1_simple_report_same_partner
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 564-567

Verbatim source input

```
LEMMA 2. If P'(mi) is a simple misrepresentation such that y = g(P') and either
y(m)P(mi)x(mi) or y(mi)= x(mi) then, for each mj in M, either y(mj)P(mj)x(mj) or

y(mj) = x(mj).
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
  [inst_3 : DecidableEq W] (val_m : M → W → ℝ) (val_w : W → M → ℝ) (m : M) (simple_report_m : W → ℝ) (ystar : W),
  Fintype.card M = Fintype.card W →
    Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
      Roth82StableMatching.PaperInterface.strictMarriageDomain (Function.update val_m m simple_report_m) val_w →
        (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
                  val_w).m_match
              m =
            some ystar →
          ((match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
              | none => 0
              | some w => val_m m w) ≤
              match
                (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
                      val_w).m_match
                  m with
              | none => 0
              | some w => val_m m w) →
            Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst simple_report_m ystar →
              ∀ (m' : M),
                (match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m' with
                  | none => 0
                  | some w => val_m m' w) ≤
                  match
                    (Roth82StableMatching.PaperInterface.menDeferredAcceptance (Function.update val_m m simple_report_m)
                          val_w).m_match
                      m' with
                  | none => 0
                  | some w => val_m m' w
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.lemma2_simple_report_harms_no_man
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 678-684

Verbatim source input

```
THEOREM 6. No feasible outcome is strictly preferred by all mi in M to the outcome g(P).

The fact that no stable outcome is strictly preferred to g(P) isn't news: Theorem 2
gives a stronger result. What Theorem 6 says is that in fact, g(P) is weakly Pareto
optimal from the point of view of the men, with respect to any possible outcome. The

example above shows that this can't be strengthened to strong Pareto optimality.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] [inst_2 : DecidableEq M]
  [inst_3 : DecidableEq W] [Nonempty M] (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Fintype.card M = Fintype.card W →
    Roth82StableMatching.PaperInterface.strictMarriageDomain val_m val_w →
      ¬∃ nu,
          Roth82StableMatching.PaperInterface.completeMarriageOutcome nu ∧
            ∀ (m : M),
              (match (Roth82StableMatching.PaperInterface.menDeferredAcceptance val_m val_w).m_match m with
                | none => 0
                | some w => val_m m w) <
                match nu.m_match m with
                | none => 0
                | some w => val_m m w
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.theorem6_no_outcome_strictly_better_for_all_men
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 713-714

Verbatim source input

```
THEOREM 7. No stable matching procedure exists which never gives any agent an
incentive to misrepresent his kth choice, for k -# 1.
```

Expanded PaperInterface specification

```
∀ (k : ℕ),
  1 < k → ∃ n, Roth82StableMatching.paper_no_stable_procedure_avoids_strict_kth_choice_manipulation_on (Fin n) (Fin n) k
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.theorem7_later_choice_manipulation_unavoidable
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 18

Source locator: cited publication, lines 522-529

Verbatim source input

```
We first show that only a certain kind of simple misrepresentation need

considered, since if any successful manipulation is possible, then it can be achieved
a simple misrepresentation. Specifically, if P'(mi) is an arbitrary misrepresentation
discussed above, then an equivalent simple misrepresentation P"(mi) is one in whic

y(mi)P"(mi)wj for all wj #y(mi). That is, P"(mi) is a preference relation which ha
y(m,) as the most preferred match of mi,. The justification for calling P"(mi) an
```

Expanded PaperInterface specification

```
∀ {W : Type u_1} (report_m : W → ℝ) (wstar : W),
  Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst report_m wstar ↔
    ∀ (w : W), w ≠ wstar → report_m w < report_m wstar
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.manReportStrictlyRanksPartnerFirst_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 300-303

Verbatim source input

```
4. Incentives and stability. Since each agent alone knows his own preferences, any
matching procedure which depends on agents' preferences can be thought of as
consisting of two parts: a mechanism for eliciting the preferences of the agents, and a
mechanism for aggregating these elicited preferences to determine an outcome. This
```

Expanded PaperInterface specification

```
∀ (M : Type u) (W : Type v),
  Roth82StableMatching.PaperInterface.matchingProcedure M W =
    (Roth82StableMatching.PaperInterface.preferenceProfile M W →
      { mu // Roth82StableMatching.PaperInterface.completeMarriageOutcome mu })
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.matchingProcedure_eq_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 463-466

Verbatim source input

```
results, which make use of the fact that Theorem 1 permits us to identify, for each set

of agents, a unique optimal stable outcome, which they each like at least as well as any
other stable outcome.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} (val_m : M → W → ℝ) (val_w : W → M → ℝ) (mu : EconCSLib.Matching.Assignment M W),
  (Roth82StableMatching.PaperInterface.sourceMenOptimal val_m val_w mu ↔
      Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu ∧
        ∀ (mu' : EconCSLib.Matching.Assignment M W),
          Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu' →
            ∀ (m : M),
              (match mu'.m_match m with
                | none => 0
                | some w => val_m m w) ≤
                match mu.m_match m with
                | none => 0
                | some w => val_m m w) ∧
    (Roth82StableMatching.PaperInterface.sourceWomenOptimal val_m val_w mu ↔
      Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu ∧
        ∀ (mu' : EconCSLib.Matching.Assignment M W),
          Roth82StableMatching.PaperInterface.sourceStableMarriage val_m val_w mu' →
            ∀ (w : W),
              (match mu'.w_match w with
                | none => 0
                | some m => val_w w m) ≤
                match mu.w_match w with
                | none => 0
                | some m => val_w w m)
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.sideOptimalStableOutcome_iff_source_definition
```

Recorded source-to-Spec screening Verdict: `not recorded`

Reason: not recorded
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 21

Source locator: cited publication, lines 176-197

Verbatim source input

```
the marriage problem was defined as having equal numbers of men and women, and
without the possibility that a feasible outcome could leave any agent unmatched, while
in the general matching problem, there may be an excess of one kind of agent, and an

outcome need not make a match for every agent. This more general case can be
handled by adding suitable "dummy" agents corresponding to the option of being
unmatched in the final outcome. The third respect in which the marriage problem as
described above is more restrictive than the general matching problem is that indiffer-

ence between potential partners has been ruled out by the assumption that all
preferences are strict. Relaxing this assumption would actually complicate some of the
results. To avoid these complications, only strict preferences will be considered.

Thus the marriage problem will be used to represent the general matching problem
with strict preferences. This should not obscure the fact that much of the interest of the
5That is, the preference relations P(m,) (or P(wi)) have the following properties:
(1) transitivity: if WjPiWk and wkPiwI, then wjPiwl;

(2) completeness: for all distinct wj, wk in W, exactly one of the relations wjPiWk or wkP,wj holds
The interpretation of the preference relations Pi as strict preference relations means that we are ruling out
the possibility that an agent will be indifferent between two potential partners. (Also, it is never the case that
wjP.jW.) See the discussion at the end of this section.
```

Expanded PaperInterface specification

```
∀ {M : Type u_1} {W : Type u_2} [inst : Fintype M] [inst_1 : Fintype W] (val_m : M → W → ℝ) (val_w : W → M → ℝ),
  Roth82StableMatching.PaperInterface.sourceMarriageModel val_m val_w ↔
    Fintype.card M = Fintype.card W ∧
      (∀ (m : M) (w w' : W), val_m m w = val_m m w' → w = w') ∧ ∀ (w : W) (m m' : M), val_w w m = val_w w m' → m = m'
```

Lean proof endpoint (built separately)

```
Roth82StableMatching.PaperInterface.sourceMarriageModel_iff_source_definition
```

Recorded source-to-Spec screening **Verdict:** `not recorded`

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation